Variance-retained sequential atoms, self-to-sequential homotopy, and the directed-refinement terminal boundary

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1. Purpose and scope

R-134 left two obligations adjacent but logically distinct. The first was an owner-preserving passage from the R-087 self atom to the R-088 sequential atom, with the conditional future variance and the R-063 forest retained. The second was a terminal Gaussian reservation with enough joint spatial data to support the conditional $\gamma = 7/12$ route. This note audits both obligations against R-104, R-125, and the directed source union.

The conclusions are deliberately asymmetric.

1. The R-125 future-variance rebate gives the exact owner coordinate. It must remain inside the sequential atom sum; replacing it by an uncentred second-moment envelope can erase an exact cancellation before any shell estimate is attempted.

2. R-088 gives an exact sequential physical-shell atom. The R-087 spatial proof reruns directly on that atom because its band-limited coefficient path has exactly the same three-channel integrand. This proves the uncentred far-output atom estimate, but not the centred conditioning step or the once-owned $\sum_k q_k^{\mathrm{seq}}$ bound.

3. R-087 self atoms and R-088 sequential atoms are two path decompositions of the same endpoint. An exact two-parameter homotopy identifies their termwise difference. Equality of the total sums does not give termwise domination, but that comparison is unnecessary for the direct spatial reproof; the remaining production obstruction is the centred owner conversion and summability of the sequential atom costs.

4. The extracted translated-model Z^6 route still fails on the accepted predictable rare event. The failure is an envelope no-go, not a counterexample to the exact sequential atom.

5. A nonzero Gaussian block cannot remain independent and uniformly six-real elliptic for every chart in the full R-093 directed temporal refinement union. Its terminal-tail covariance converges to zero as the last mesh interval shrinks. This is a structural no-go for the refinement-uniform last-block architecture, not merely an absent estimate.

6. The viable successor is therefore fixed-chart or shell-weighted: keep a finite unrevealed aggregate when its chart-dependent covariance is elliptic, or absorb shell-dependent inverse-covariance and spatial costs into a once-owned q_k ledger. Neither successor is proved for the complete production owners here.

The scope is one fixed finite cutoff, one positive density floor, one common legal heat, and one declared R-093/R-104 temporally faithful chart unless a statement explicitly quantifies over refinements.

All representation-preserving same-root visits are recombined. Output clusters are contraction closed, orthogonal, and exhaustive. The source energy and terminal sextic remain outside the owner identity and may be appended only once after all owners are summed.

The binding authorities are R-087, R-088, R-091, R-092, R-104, R-116, R-119, R-123, R-125, R-129, R-133, and R-134. No statement below changes their tiers or promotes R-135 beyond its registered scoped T4 authority.

2. Exact owner identity and the variance firewall

Fix a recombined root–spatial owner o. Let $\mathcal{C}$ range over its exhaustive orthogonal output family and let

$$X_o = \left(P_{\mathcal{C}} C_{6,e_{o,\mathcal{C}}}(W_{o,\mathcal{C}}) V_{o,\mathcal{C}}\right)_{\mathcal{C}} \in \bigoplus_{\mathcal{C}} \mathcal{H}_{\mathcal{C}}. \tag{2.1}$$

Condition on the retained legal data and integrate only the declared future; write this conditional expectation as $\mathbb{E}_f$. Put

$$\Phi_o = \mathbb{E}_f X_o, \qquad \mathcal{V}_o = \mathbb{E}_f \|X_o - \Phi_o\|^2. \tag{2.2}$$

R-125 conditional Pythagoras and the once-owned trace probe give

$$\mathbb{E}_f \|X_o\|^2 = \|\Phi_o\|^2 + \mathcal{V}_o. \tag{2.3}$$

After the complete visit telescope and the adapted forest reconstruction, the compact owner identity is

$$\boxed{\mathcal{P}_{\mathrm{comp},o} = \frac{1}{2}\mathrm{F}^{\mathrm{ad}}_{063,o} - \frac{1}{2}\operatorname{Var}_f(X_o).} \tag{2.4}$$

Equivalently, for the moving endpoint,

$$\boxed{\Psi_o = \Delta\mathcal{V}_o - \Delta\mathrm{F}^{\mathrm{ad}}_{063,o}.} \tag{2.5}$$

The factor 1/2 in (2.4) is load bearing. The cluster direct sum must be formed before the variance is taken. No forest share is allocated to one cluster, and no copy of the target covariance is allocated to one visit.

The elementary identity

$$\operatorname{Var}_f(X_o) = \mathbb{E}_f \|X_o\|^2 - \|\mathbb{E}_f X_o\|^2 \tag{2.6}$$

exhibits the first rebate. Replacing the variance by the uncentred second moment gives the weaker lower bound

$$\mathcal{P}_{\mathrm{comp},o} \geq \frac{1}{2}\mathrm{F}^{\mathrm{ad}}_{063,o} - \frac{1}{2}\mathbb{E}_f \|X_o\|^2, \tag{2.7}$$

and discards the positive term $\|\mathbb{E}_f X_o\|^2/2$. That discarded term is not a spare reserve: it is the conditional-mean coordinate paired with the primitive trace in the exact R-125 bridge.

Let $\mathcal{O}_{\mathrm{full}}$ include every recombined root owner together with the complete-low/stationary owner required by R-123, and let Θ_o denote the once-owned primitive trace paired with Φ_o. With

$$\mathfrak{F}_{\mathrm{full}} = \sum_{o\in\mathcal{O}_{\mathrm{full}}} \mathbb{E}\mathrm{F}^{\mathrm{ad}}_{063,o}, \qquad \mathfrak{V}_{e,\mathrm{full}} = \sum_{o\in\mathcal{O}_{\mathrm{full}}} \mathbb{E}\operatorname{Var}_f(X_o),$$

the exact aggregate coordinate is

$$\boxed{2\sum_{o\in\mathcal{O}_{\mathrm{full}}} \mathbb{E}\mathcal{P}_{\mathrm{comp},o} = \mathfrak{F}_{\mathrm{full}} - \mathfrak{V}_{e,\mathrm{full}}.} \tag{2.8}$$

The minimal live target is therefore the R-123 trace-excess estimate

$$\boxed{\sum_{o\in\mathcal{O}_{\rm full}} \mathbb{E}\{\Theta_o - \|\Phi_o\|^2\} \le 2\eta X_{\rm src} + 2\zeta Y_6 + 2C,} \qquad \eta < \frac{9}{20}, \quad \zeta < \frac{3}{20}. \tag{2.9}$$

Only after the full owner aggregation may this be written equivalently as

$$\boxed{\mathfrak{F}_{\rm full} - \mathfrak{V}_{e,\rm full} \ge -2\eta X_{\rm src} - 2\zeta Y_6 - 2C.} \tag{2.10}$$

The action reserve would then be

$$\boxed{\mathcal{A}_J(h) \ge \left(\frac{9}{20} - \eta\right) X_{\rm src} + \left(\frac{3}{20} - \zeta\right) Y_6 - C,} \tag{2.11}$$

with only the nonnegative R-104 Douglas slack added in the physical progressive action. A floor split must retain $\operatorname{Var}_f(X_0 + R_e) = \operatorname{Var}_f X_0 + 2\operatorname{Cov}_f(X_0, R_e) + \operatorname{Var}_f R_e$. Silently absorbing the low owner into the root-only R-063 symbol changes the meaning of the established forest and is not allowed.

3. Envelope–rebate erasure counterfixtures

3.1 The primitive-trace fixture

Take trivial past, one future $\eta \sim N(0,1)$, one full output cluster, no heat, base $w_0 = 0$, terminal $w_1 = e_1$, and derivative

$$V = \eta e_2, \qquad \Gamma = e_2 e_2^T. \tag{3.1}$$

With

$$s = c_0 + c_1 = \frac{339}{8000P}, \tag{3.2}$$

the complete six-row calculation is

$$J_1 = 2\eta(\sqrt{c_0}, \sqrt{c_1}), \qquad \Phi_1 = 0, \qquad \Theta_1 = 4s, \qquad \mathcal{V}_1 = 4s. \tag{3.3}$$

The covariance-normal endpoint and its deterministic R-063 forest both have mean zero:

$$\mathbb{E}\{2s(\eta^2 - 1)\} = 0. \tag{3.4}$$

The correct bridge reads $0 = (4s - 4s)/2$. Deleting the conditional variance would give the false value $-2s$ and miss exactly

$$2s = \frac{339}{4000P}. \tag{3.5}$$

Thus even a deterministic translation needs the variance rebate. This is a counterfixture to rebate erasure, not to (2.4).

3.2 The scaled R-125 scalar-envelope fixture

Scale the exact one-owner, one-exhaustive-cluster R-125 fixture by $\nu > 0$:

$$\beta_{\rm op} = 4(c_0 + c_1) = \frac{339}{2000P}. \tag{3.6a}$$

$$W = e_1, \qquad V_\nu = \nu\eta e_2, \qquad \Gamma_\nu = \nu^2 e_2 e_2^T. \tag{3.6}$$

The six-row Pauli frame gives

$$X_\nu = 2\nu\eta(\sqrt{c_0}, \sqrt{c_1}), \quad \Phi_\nu = 0, \quad \Theta_\nu = \operatorname{Var}_f(X_\nu) = \beta_{\rm op}\nu^2, \tag{3.7}$$

and hence

$$\mathbb{E}\mathrm{F}^{\mathrm{ad}}_{063} = 0, \qquad \mathbb{E}\mathcal{P}_{\mathrm{comp}} = -\frac{\beta_{\mathrm{op}}}{2}\nu^2. \tag{3.8}$$

Equivalently, the exact R-123 coordinate on this fixture is

$$\mathbb{E}\{\Theta_\nu - \|\Phi_\nu\|^2\} = \beta_{\mathrm{op}}\nu^2 = -2\mathbb{E}\mathcal{P}_{\mathrm{comp}}. \tag{3.8a}$$

For the R-134 scalar envelope,

$$A^2 = \nu^2, \qquad B_e^2 = e\nu^2, \tag{3.9}$$

and hence $Q_e = \nu^2(\sqrt{339/(2000P)} + \sqrt{3e/(320P)})^2$, so that

$$\boxed{\mathfrak{F} - Q_e = -\nu^2\left(\sqrt{\frac{339}{2000P}} + \sqrt{\frac{3e}{320P}}\right)^2 \longrightarrow -\infty.} \tag{3.10}$$

Thus R-134's valid lower surrogate cannot itself be uniformly lower bounded from the root-owner scalar norms. This refutes that successor architecture, not the complete torus production action: the fixture does not impose the full spatial law or include the complete-low/stationary owner which appears in (2.8)–(2.10).

3.3 The completed-square trace fixture

Let

$$A_\eta = B_1 + 2\eta\mathrm{I}, \quad K_\eta = LA_\eta^{-1}L, \quad M_\eta = L - K_\eta, \quad Q = GG^T - \Gamma, \quad D_\eta = \frac{1}{2}K_\eta : \Gamma. \tag{3.11}$$

R-088 records the pointwise null identity

$$\boxed{\frac{1}{2}G^TK_\eta G - \frac{1}{2}K_\eta : Q - D_\eta = 0.} \tag{3.12}$$

It is the same firewall in completed-square coordinates: the square variance, Wick contraction, and trace debt are one packet.

For a fixed target dimension six, take

$$B_1 = 0, \quad L = \ell\mathrm{I}_6, \quad c = 0, \quad \Gamma_N = \rho_N\mathrm{I}_6, \quad \mathbb{E}G = 0, \quad \mathrm{Cov}(G) = \Gamma_N, \quad \rho_N \uparrow \infty. \tag{3.13}$$

The original packet has $\mathbb{E}\mathcal{P} = 0$, whereas the isolated debt is

$$D_{\eta,N} = \frac{\ell^2}{4\eta}\operatorname{Tr}\Gamma_N = \frac{6\rho_N\ell^2}{4\eta}. \tag{3.14}$$

The expected retained-square variance equals (3.14) exactly. Any absolute envelope which keeps the debt and erases its variance companion manufactures a cutoff-divergent loss from a zero-mean packet.

4. Exact sequential Cartan atom

Fix a production generator A and probability root j. Write

$$\Phi = \Phi_{A,j}, \qquad \mathcal{C}_\Phi(z) = \Phi(z)^TDz, \tag{4.1}$$

and let the physical control shells be

$$A^{(k)} = A^{(\ell)} + \sum_{r=j_0}^{k} a_r, \qquad B_{j,k} = w_j + A^{(k-1)}. \tag{4.2}$$

The sequential endpoint increment is

$$\Theta_{A,j,k} = \mathcal{C}_\Phi(B_{j,k} + a_k) - \mathcal{C}_\Phi(B_{j,k}), \tag{4.3}$$

and telescopes exactly:

$$\sum_{k=j_0}^{j} \Theta_{A,j,k} = \mathcal{C}_\Phi(w_j + A^{(j)}) - \mathcal{C}_\Phi(w_j + A^{(\ell)}). \tag{4.4}$$

Since a_k is strict-past for the fresh root j, its fresh-root derivative vanishes. Differentiating (4.3) in a fresh Cameron–Martin direction v gives the complete three-channel atom

$$\boxed{\begin{aligned} T_{A,j,k}[v] = \{&D\Phi(B_{j,k} + a_k) - D\Phi(B_{j,k})\}[v]^T DB_{j,k} \\ &+ D\Phi(B_{j,k} + a_k)[v]^T Da_k \\ &+ \{\Phi(B_{j,k} + a_k) - \Phi(B_{j,k})\}^T Dv. \end{aligned}} \tag{4.5}$$

Equivalently,

$$\begin{aligned} T_{A,j,k}[v] = \int_0^1 \{&D^2\Phi(B_{j,k} + ra_k)[a_k, v]^T D(B_{j,k} + ra_k) \\ &+ D\Phi(B_{j,k} + ra_k)[v]^T Da_k \\ &+ D\Phi(B_{j,k} + ra_k)[a_k]^T Dv\}\, dr. \end{aligned} \tag{4.6}$$

Therefore

$$\sum_{k=j_0}^{j} T_{A,j,k} = \mathbb{D}_j\{\mathcal{C}_\Phi(w_j + A^{(j)}) - \mathcal{C}_\Phi(w_j + A^{(\ell)})\}. \tag{4.7}$$

Every future physical control shell is absent from the coefficient path of the kth atom. With the R-081 bounds $\|D\Phi\|_\infty \le 3$ and $\|D^2\Phi\|_\infty \le M_{e,2}$,

$$|T_{A,j,k}[v]| \le M_{e,2}|a_k||v||DB_{j,k}| + 3|v||Da_k| + 3|a_k||Dv|. \tag{4.8}$$

For a root basis satisfying

$$\sum_h |v_h|^2 \le \kappa_0 2^{-j}, \qquad \sum_h |Dv_h|^2 \le \kappa_1 2^j, \tag{4.9}$$

one obtains the pointwise envelope

$$\sum_h |T_{A,j,k}[v_h]|^2 \le C_e\{2^{-j}|a_k|^2|DB_{j,k}|^2 + 2^{-j}|Da_k|^2 + 2^j|a_k|^2\}. \tag{4.10}$$

Equations (4.5)–(4.10) are exact or fixed-floor local estimates. They do not by themselves make distinct k atoms orthogonal.

4A. Direct transfer of the R-087 spatial theorem

The coefficient path

$$u^{\mathrm{seq}}_{j,k,r} = B_{j,k} + ra_k \tag{4A.1}$$

is Q_j-supported, and the three terms in (4.6) are exactly the three R-087 remainder channels. The R-087 proof therefore applies to the sequential atom itself; no self-to-sequential square comparison is used. For $1/3 < \alpha < 1/2$ and $1/2 < s < 3\alpha - 1/2$, define

$$W_k^{\mathrm{seq}} = \sup_{j \ge k} \sum_A \mathbb{E} \int_0^1 \big(1 + \|B_{j,k} + ra_k\|_{\mathcal{C}^\alpha}\big)^6 \big(\|a_k\|_2^2 + \|Da_k\|_2^2\big)\, dr, \tag{4A.2}$$

$$q_k^{\mathrm{seq}} = C_e 2^{-(6\alpha-1)k} W_k^{\mathrm{seq}}. \tag{4A.3}$$

Theorem 4A.1 (sequential Cartan spatial atom)

For every $m \ge j+5$,

$$\boxed{\sum_A \int_0^\infty e^{-2t}\mathbb{E}\|\Pi_m P_t^{(j)} T_{A,j,k}\|_{\mathrm{HS}}^2 dt \le 2^{-2s(m-k)} q_k^{\mathrm{seq}}.} \tag{4A.4}$$

Indeed, Jensen in r, conditional OU contraction, and the R-087 paraproduct estimates leave the positive exponent margins

$$(6\alpha - 1) - 2s, \qquad 4\alpha - 2s, \qquad 6\alpha - 2s. \tag{4A.5}$$

At $\alpha = 2/5$ and $s = 7/12$, the two critical margins are 7/30 and 13/30. At the R-134 aggregate exponent $s = 2/3$, all three margins are

$$\frac{1}{15}, \qquad \frac{4}{15}, \qquad \frac{16}{15}, \tag{4A.6}$$

respectively. This proves the uncentred spatial estimate only. It neither proves $\sum_k q_k^{\mathrm{seq}} < \infty$ nor identifies the complete centred R-125 owner current with the projected atom family.

5. Variance-retained sequential summation

Let $\mathcal{Y}_{A,j,k,t}$ be the exhaustive output-cluster direct sum of the legal projection of $P_t^{(j)} T_{A,j,k}$. Define its conditionally centred form

$$\widetilde{\mathcal{Y}}_{A,j,k,t} = \mathcal{Y}_{A,j,k,t} - \mathbb{E}_f \mathcal{Y}_{A,j,k,t}. \tag{5.1}$$

Whenever the deterministic spatial and OU operators commute with the legal future conditioning,

$$\mathbb{E}\|\Pi_m \widetilde{\mathcal{Y}}_{A,j,k,t}\|^2 \le \mathbb{E}\|\Pi_m \mathcal{Y}_{A,j,k,t}\|^2. \tag{5.2}$$

If that commutation has not been established, the centred estimate below must be assumed directly.

Theorem 5.1 (conditional variance sequential atom theorem)

Assume that for some $s > 0$, every $k \le j$, and every $m \ge j+5$,

$$\sum_A \int_0^\infty e^{-2t}\mathbb{E}\|\Pi_m \widetilde{\mathcal{Y}}_{A,j,k,t}\|^2\, dt \le 2^{-2s(m-k)} q_k, \tag{5.3}$$

where $q_k \ge 0$ is deterministic and

$$\sum_k q_k \le Q. \tag{5.4}$$

For $0 < \gamma < s$, define the variance-retained output ledger

$$\mathcal{B}_\gamma^{\mathrm{var}} := \sum_{A,j} \sum_{m \ge j+5} 2^{2\gamma(m-j-5)} \int_0^\infty e^{-2t}\mathbb{E}\,\mathrm{Var}_f\left(\Pi_m \sum_{k \le j} \mathcal{Y}_{A,j,k,t}\right) dt. \tag{5.5}$$

Then

$$\boxed{\mathcal{B}_\gamma^{\mathrm{var}} \le \frac{2^{-10s}}{(1-2^{-s})^2(1-2^{-2(s-\gamma)})} Q.} \tag{5.6}$$

Proof. Conditional Pythagoras gives

$$\mathrm{Var}_f\left(\Pi_m \sum_{k \le j} \mathcal{Y}_k\right) = \mathbb{E}_f \left\|\Pi_m \sum_{k \le j} \widetilde{\mathcal{Y}}_k\right\|^2. \tag{5.7}$$

For fixed j, m, t, weighted Cauchy–Schwarz gives

$$\left\| \sum_{k\le j} \Pi_m \widetilde{\mathcal{Y}}_k \right\|^2 \le \frac{1}{1-2^{-s}} \sum_{k\le j} 2^{s(j-k)} \|\Pi_m \widetilde{\mathcal{Y}}_k\|^2. \tag{5.8}$$

Insert (5.3), use Tonelli, and put $r = j-k$ and $d = m-j \ge 5$. The r series contributes $(1-2^{-s})^{-1}$, while

$$\sum_{d\ge 5} 2^{2\gamma(d-5)} 2^{-2sd} = \frac{2^{-10s}}{1-2^{-2(s-\gamma)}}. \tag{5.9}$$

Equations (5.4), (5.8), and (5.9) prove (5.6). ☐

At

$$s = \frac{2}{3}, \qquad \gamma = \frac{7}{12}, \tag{5.10}$$

the square constant is

$$\boxed{C^{\mathrm{var}}_{2/3,7/12} = \frac{2^{-20/3}}{(1-2^{-2/3})^2(1-2^{-1/6})} = 0.658881625872614\ldots.} \tag{5.11}$$

No cross-k orthogonality is used, and each q_k is spent once. The theorem does not prove (5.4) or the identification of the complete R-125 owner current with the projected R-088 atom sum. Theorem 4A.1 supplies the uncentred spatial version of (5.3) with $q_k = q_k^{\mathrm{seq}}$. The centred version follows from (5.2) only after the legal conditioning commutation is certified; otherwise it remains a separate production input. The once-owned q_k^{seq} sum and owner intertwiner remain open.

6. Self atoms versus sequential atoms: an exact homotopy

R-087 and R-088 atomise the same endpoint along different paths. The difference can be located without guessing a sign.

Let E and H be finite-dimensional real Hilbert spaces, let $F : E \to H$ be C^2, and let

$$a = \sum_{k=1}^{K} a_k, \qquad p_{k-1} = \sum_{r<k} a_r. \tag{6.1}$$

Define the straight-path self atoms and polygonal sequential atoms by

$$M_k = \int_0^1 DF(ra)[a_k]\, dr, \tag{6.2}$$

$$T_k = \int_0^1 DF(p_{k-1} + ra_k)[a_k]\, dr. \tag{6.3}$$

Both decompositions telescope:

$$\boxed{\sum_k M_k = F(a) - F(0) = \sum_k T_k.} \tag{6.4}$$

For the exact termwise difference put

$$H_k(r) = p_{k-1} + ra_k - ra = (1-r)p_{k-1} - r\sum_{q>k} a_q, \tag{6.5}$$

$$\Gamma_k(r,\tau) = ra + \tau H_k(r). \tag{6.6}$$

The fundamental theorem of calculus in τ gives

$$\boxed{T_k - M_k = \int_0^1 \int_0^1 D^2 F(\Gamma_k(r,\tau))[H_k(r), a_k]\, d\tau\, dr.} \tag{6.7}$$

Summing (6.7) gives zero by (6.4), but no summand has a fixed sign.

For the production comparison take

$$F(A) = \mathbb{D}_j \mathcal{C}_{\Phi_{A,j}}(w_j + A), \tag{6.8}$$

with the finite-low shift absorbed into the origin. Then (6.2) is the R-087 self atom and (6.3) is the R-088 sequential atom. Equation (6.5) shows the precise trade: the sequential path removes future controls from its kth coefficient, while the self-to-sequential homotopy contains past-prefix and future-tail mixed Hessian terms. Consequently

$$\sum_k \|T_k\|^2 \lesssim \sum_k \|M_k\|^2 \tag{6.9}$$

does not follow from endpoint exactness. A proof of (6.9), a weighted variant, or any later attempt to compare the two decompositions must estimate the strip (6.7) with the R-125 variance and forest owners retained. Theorem 4A.1 does not require (6.9): it proves the sequential spatial estimate directly from the common three-channel integrand. The remaining burden is to sum its actual q_k^{seq} costs and intertwine the legally centred full owners.

7. Extracted-Z^6 rare-event no-go

The R-087 straight-path spatial estimate introduced

$$q_k^{\mathrm{mod}} = C_e 2^{-(6\alpha-1)k} \sup_{j\ge k} \sum_A \mathbb{E} \int_0^1 Z_{j,r}^6 \{\|a_k\|_2^2 + \|Da_k\|_2^2\}\, dr. \tag{7.1}$$

The R-088 direct-root theorem needs only $\sum_k q_k$, but this particular extraction still fails. Put $N = 2^k$, choose $E \in \mathcal{F}_{k-1}$ with $\mathbb{P}(E) = p = N^{-6}$, and set

$$h_k = N^3 \mathbf{1}_E e_N, \qquad a_k = K_k h_k = N \mathbf{1}_E e_N. \tag{7.2}$$

Then

$$\mathbb{E}X \asymp \mathbb{E}Y \asymp \mathbb{E}\sqrt{XY} \asymp 1, \tag{7.3}$$

whereas, with $\beta = 6\alpha - 1$,

$$\boxed{q_k^{\mathrm{mod}} \asymp N^{-\beta} p N^{6(1+\alpha)} N^4 = N^5 = p^{-5/6}.} \tag{7.4}$$

The older 2^k-weighted extraction grows as $N^6 = p^{-1}$. Thus neither the weighted nor unweighted theorem converts this Z^6 envelope into a one-use ledger.

This fixture does not evaluate the exact atom (4.5). Indeed, the exact scalar production saturation analysed in R-091 has a fixed-gap quantity of order N^{-4} on the same rare event. Therefore (7.4) proves only

$$\text{extracted translated-model } Z^6 \text{ majorants are too coarse;} \tag{7.5}$$

it neither falsifies nor proves the expectation-inside sequential estimate (5.3).

8. Conditioning, rank, and spatial regularity

For a jointly Gaussian finite-dimensional pair (Z, Y),

$$\boxed{\operatorname{Cov}(Z \mid Y) = \Gamma_Z - \Gamma_{ZY}\Gamma_Y^{\dagger}\Gamma_{YZ}.} \tag{8.1}$$

If $Y = AZ$, then

$$\operatorname{Cov}(Z \mid AZ) = \Gamma_Z^{1/2}(\mathrm{I} - P_{\operatorname{Ran}(\Gamma_Z^{1/2}A^*)})\Gamma_Z^{1/2}. \tag{8.2}$$

Thus independent conditioning preserves covariance, while noiseless observation can remove rank, including all of it when $A = \mathrm{I}$.

For a Gaussian field ζ with covariance kernel K, a finite Gaussian observation Y gives

$$\mathbb{E}[\zeta(x) \mid Y] = m(x) + K_{\zeta Y}(x)K_Y^{\dagger}(Y - \mathbb{E}Y), \tag{8.3}$$

$$K_{|Y}(x, y) = K(x, y) - K_{\zeta Y}(x)K_Y^{\dagger}K_{Y\zeta}(y). \tag{8.4}$$

Consequently

$$\operatorname{Cov}(\partial_i\zeta(x) \mid Y) = \partial_{x_i}\partial_{y_i}K_{|Y}(x, y)|_{y=x}, \tag{8.5}$$

$$\mathbb{E}[|\zeta(x) - \zeta(y)|^2 \mid Y] = \operatorname{Tr}\{K_{|Y}(x, x) + K_{|Y}(y, y) - 2K_{|Y}(x, y)\}. \tag{8.6}$$

The residual covariance decreases as a quadratic form, but its smallest eigenvalue can collapse and the regression mean contains the pseudoinverse $K_Y^{\dagger}$. Conditioning on a nonlinear same-root coefficient need not leave a Gaussian conditional law, so (8.1)–(8.6) then require a separate affine representation theorem.

The sharp spatial diagnostic is

$$\zeta_N^a(x) = G_a\cos(Nx) + H_a\sin(Nx), \qquad a = 1, \ldots, 6, \tag{8.7}$$

with independent standard G_a, H_a. It has

$$K_N(x, y) = \cos(N(x - y))\mathrm{I}_6, \qquad K_N(x, x) = \mathrm{I}_6, \tag{8.8}$$

but its $W^{\sigma,3}$ norm grows as N^{σ}. After conditioning on $G = (G_a)$,

$$\mathbb{E}[\zeta_N(x) \mid G] = G\cos(Nx), \qquad \operatorname{Cov}(\zeta_N(x) \mid G) = \sin^2(Nx)\mathrm{I}_6, \tag{8.9}$$

$$\operatorname{Cov}(\partial_x\zeta_N(x) \mid G) = N^2\cos^2(Nx)\mathrm{I}_6. \tag{8.10}$$

Hence $\inf_x \lambda_{\min} = 0$ after conditioning despite exact unit pointwise covariance before conditioning. R-116 additionally forbids splitting the production sine/cosine and polarization partners into an artificial standalone root; they must remain in one legal conditional cluster. The complete pair still proves that pointwise ellipticity alone does not imply a spatial fractional estimate.

9. Directed-refinement terminal no-go

R-104 starts from a finite temporal partition and defines

$$\Delta C_k = \int_{I_k} J_tJ_t^*\,dt, \qquad S_k = \Delta C_k^{1/2}, \qquad X_J \overset{\text{law}}{=} \sum_k S_k\xi_k. \tag{9.1}$$

Singular ΔC_k are allowed. Let a refinement have final cut $s < 1$. The Gaussian source component independent of $\mathcal{F}_s$ and contained in the physical terminal source is

$$Y_s = \int_s^1 J_t\,dW_t, \qquad C_s = \operatorname{Cov}(Y_s) = \int_s^1 J_tJ_t^*\,dt. \tag{9.2}$$

Let $E_x : H_J \to \mathbb{R}^6$ be point evaluation at fixed cutoff and put

$$\Gamma_s(x) = E_xC_sE_x^*. \tag{9.3}$$

Theorem 9.1 (no refinement-uniform last innovation)

For every fixed cutoff and point,

$$\boxed{\lambda_{\min}\Gamma_s(x) \le \frac{1}{6}\operatorname{Tr}\Gamma_s(x) = \frac{1}{6}\int_s^1 \|J_t^* E_x^*\|_{\mathrm{HS}}^2\,dt \longrightarrow 0 \quad (s \uparrow 1).} \tag{9.4}$$

Therefore no $\lambda_* > 0$ can satisfy

$$\Gamma_s(x) \succeq \lambda_* \mathrm{I}_6 \tag{9.5}$$

uniformly in the R-093 directed temporal refinements. If the integrand in (9.4) is bounded by $M_{J,x}^2$, then

$$\lambda_{\min}\Gamma_s(x) \le \frac{1-s}{6} M_{J,x}^2. \tag{9.6}$$

Proof. The trace bound is elementary. The final integral tends to zero by absolute continuity of the finite-cutoff covariance integral. □

There is also an invariant Gaussian-Hilbert-space formulation. A Gaussian linear functional independent of every $\mathcal{F}_s$, $s < 1$, must have a source kernel supported in

$$\bigcap_{s<1} (s, 1] = \{1\}, \tag{9.7}$$

which is null for the time covariance measure; the functional is therefore zero in L^2. Freezing a positive-length last interval avoids (9.4) only by removing late predictable controls and is not cofinal in the full progressive core.

The increasing sharp-cube root filtration gives the same obstruction in discrete form. The accepted production value-root estimate is

$$\operatorname{Tr}\Sigma_j^{\mathrm{val}}(x) \le \kappa_0 2^{-j}. \tag{9.8}$$

For the future tail

$$\Gamma_{>j}^{\mathrm{val}}(x) = \sum_{r>j} \Sigma_r^{\mathrm{val}}(x),$$

one has

$$\lambda_{\min}\Gamma_{>j}^{\mathrm{val}}(x) \le \frac{\kappa_0}{6} 2^{-j}, \qquad \Gamma_{>J}^{\mathrm{val}} = 0. \tag{9.9}$$

Moreover, for a centred packet Z_j and $q > 0$, Jensen gives

$$\mathbb{E}|Z_j|^{-q} \ge (\mathbb{E}|Z_j|^2)^{-q/2}. \tag{9.10}$$

At the R-134 inverse orders $q = 9/4$ and $q = 9/2$, a value covariance of scale 2^{-j} therefore carries growth $2^{9j/8}$ and $2^{9j/4}$, respectively. The sharp R-134 upper constants are

$$C_{6,9/4} = \frac{\Gamma(15/8)}{2^{17/8}}, \qquad C_{6,9/2} = \frac{\Gamma(3/4)}{2^{13/4}}. \tag{9.11}$$

Full rank at each fixed j cannot turn these shell-dependent constants into an owner-uniform one.

Finally, the current R-125 conditioning retains the current root and integrates only legal future variables. For the maximal root there is no future block, so its future conditional covariance is zero. Reintegrating that retained root, reordering a low block to the end, or adding a fresh independent coordinate changes the owner disintegration or source chart and requires a new theorem. R-104 explicitly claims no invariance under such a change of reveal order or fresh coordinates.

10. Fixed-chart and weighted successors

10.1 Fixed-chart aggregate packet

For one declared chart choose a tail set T which remains unrevealed at the conditioning under consideration and put

$$\zeta_T = \sum_{r\in T} S_r \xi_r, \qquad \Gamma_T(x,y) = \sum_{r\in T} E_x S_r S_r^* E_y^*. \tag{10.1}$$

An aggregate may restore rank even when its summands are singular, since

$$\operatorname{Ker}\left(\sum_{r\in T} S_r S_r^*\right) = \bigcap_{r\in T} \operatorname{Ker} S_r^*. \tag{10.2}$$

Define the chart-dependent quantities

$$\lambda_{J,\pi,T} = \inf_x \lambda_{\min}\Gamma_T(x,x), \tag{10.3}$$

$$Q_{J,\pi,T} = \sum_o \mathbb{E}\,\mathfrak{R}_o(\zeta_T)^2, \tag{10.4}$$

where the exact R-092/R-134 functional is

$$\mathfrak{R}_o = N_{+,o}\|Da_o\|_6 + \{D_{\sigma,o} + D_{\infty,o}(N_{+,o} + N_{-,o})\}\|Dx_o\|_6. \tag{10.5}$$

If $\lambda_{J,\pi,T} > 0$, the packet is independent of the conditioning, the required value–gradient regression and Fubini/Jensen bounds hold, and (10.4) is finite, then the R-134 fractional theorem applies on that fixed chart. Its second- and third-jet bounds are

$$\mathbb{E}\|D^2F_S\|^2 \le \frac{49}{\lambda_{J,\pi,T}}, \qquad \mathbb{E}\|D^3F_S\|^2 \le \frac{3249}{2\lambda_{J,\pi,T}^2}. \tag{10.6}$$

This construction covers only owners whose legal sigma-fields precede the whole set T. It is neither an every-owner construction nor stable under directed refinement.

10.2 Weighted production successor

The refinement-stable replacement is not a common positive lower bound. It is a once-owned family q_k which absorbs the actual shell covariance, value–gradient regression, spatial increments, owner conversion, and the centred sequential atom. The exact sufficient target is (5.3)–(5.4), or a direct integrated estimate of (5.5), with

$$s > \frac{7}{12}. \tag{10.7}$$

A natural shell-standardised input would have the form

$$c_- 2^{-k} \mathrm{I}_6 \preceq \Gamma_k(x,x) \preceq c_+ 2^{-k}\mathrm{I}_6, \tag{10.8}$$

$$\mathbb{E}|2^{k/2}(\zeta_k(x) - \zeta_k(y))|^r \le C_r \min\{1, (2^k|x-y|)^r\}, \tag{10.9}$$

together with a uniform standardised value–gradient regression estimate. Equations (10.8)–(10.9) are a proposed sufficient data package, not a theorem of this note. The lower matrix bound, the owner intertwiner, and the legally centred sequential owner estimate must be proved from the production symbol without splitting the legal quadrature/polarisation cluster; the uncentred spatial atom estimate itself is already Theorem 4A.1.

11. Conditional collar headroom

Suppose the variance-retained sequential theorem supplies a one-use Q and let

$$C_{\mathrm{var}} = C^{\mathrm{var}}_{2/3,7/12}. \tag{11.1}$$

After the exact R-091 output-gap extraction, an owner conversion constant Λ_{own} gives the conditional tail amplitude

$$a_{\mathrm{tail}}(C) \le M 2^{-7(C-5)/12}, \qquad M = \Lambda_{\mathrm{own}} \sqrt{C_{\mathrm{var}} Q}. \tag{11.2}$$

Let the source and sextic diagonal margins be $e_0, f_0 > 0$, let a low block satisfy $D \succeq d\mathrm{I}$, and let its coupling norm be k. Put

$$\sigma_0 = \frac{k^2}{d} < \min(e_0, f_0), \tag{11.3}$$

$$a_* = 2\sqrt{(e_0 - \sigma_0)(f_0 - \sigma_0)}. \tag{11.4}$$

The exact fixed-collar acceptance condition is

$$\boxed{a_0(C) + M 2^{-7(C-5)/12} < a_*.} \tag{11.5}$$

If a separate theorem gives $a_0(C) \le a_0 < a_*$ and $h_* = a_* - a_0 > 0$, then the smallest sufficient integer is

$$C_{\min} = \max\left\{5, \left\lfloor 5 + \frac{12}{7}\log_2 \frac{M}{h_*} \right\rfloor + 1\right\}. \tag{11.6}$$

The floor-plus-one enforces strictness. Tail decay does not create the missing headroom: if

$$a_*(C) - a_0(C) = \frac{M}{2} 2^{-7(C-5)/12}, \tag{11.7}$$

then (11.5) fails for every C.

12. Route map

Route	Verdict	Exact boundary or successor
R-125 owner coordinate	exact	Forest minus conditional future variance, with the exhaustive cluster sum formed first.
R-123 aggregate target	exact coordinate, open estimate	Include complete low/stationary ownership and bound (2.9), equivalently (2.10), before spending source and sextic once.
Uncentred second moment	failed method	It erases $\|\mathbb{E}_f X\|^2/2$; the primitive fixture (3.1)–(3.5) misses exactly $339/(4000P)$.
R-134 scalar covariance envelope	failed successor	The scaled fixture (3.6)–(3.10) sends $\mathfrak{F} - Q_e$ to minus infinity.
Separated completion debt	failed method	The pointwise null identity (3.12) makes square variance and debt one packet; (3.13)–(3.14) diverges only after separation.
R-088 sequential atom	exact with spatial transfer	The three channels and endpoint telescope are exact; Theorem 4A.1 directly proves the uncentred far-output estimate, while the q_k^{seq} sum remains open.
Centred sequential summation	conditional advance	Theorem 5.1 gives the exact $0.658881625872614\ldots$ square constant under (5.3)–(5.4).
R-087 self to R-088 sequential	direct spatial proof, no comparison	The homotopy strip (6.7) has no termwise sign, but the direct three-channel reproof in Theorem 4A.1 bypasses that comparison.
Extracted Z^6 ledger	failed method	The predictable rare branch grows N^5 unweighted and N^6 weighted; it does not test the exact atom.
Uniform last source block	ruled out over the directed union	Terminal-tail covariance tends to zero by (9.4); the maximal retained owner has zero future covariance.
Fixed-chart aggregate	conditional only	Several unrevealed blocks may restore rank, with chart-dependent $\lambda_{J,\pi,T}$ and joint spatial cost.
Shell-weighted successor	open, preferred subtheorem	Prove (5.3)–(5.4) with shell-dependent covariance and joint value–gradient data inside q_k; this feeds but does not replace the complete R-123 signed trace-excess target.
Fixed-collar Schur step	conditional only	Requires (11.5), including a separately certified positive near/balanced headroom.

13. Devil's-advocate review

1. **Objection: The exact owner identity makes the complete packet nonnegative. UPHELD as an overclaim.** Equation (2.4) fixes ownership and factors. It supplies no sign for the forest-minus-variance combination.

2. **Objection: Nonnegativity of the R-125 covariance-normal endpoint closes the smaller R-123 action owner. DISMISSED.** R-129 gives $\mathcal{E}_{\mathrm{CN}} = \mathcal{P}_{\mathrm{comp}} + \mathcal{V}/2$, in the wrong direction for lower-bounding $\mathcal{P}_{\mathrm{comp}}$. On the scaled fixture $\mathbb{E}\mathcal{E}_{\mathrm{CN}} = 0$ while $\mathbb{E}\mathcal{P}_{\mathrm{comp}} = -\beta_{\mathrm{op}}\nu^2/2 < 0$.

3. **Objection: Conditional variance may be replaced by the second moment because the latter is larger. DISMISSED as a closure method.** The replacement is a formally valid weaker lower bound, but it deletes the mean rebate. Equations (3.1)–(3.5) show that the deleted amount is already load bearing in a deterministic translation.

4. **Objection: The completed-square debt is positive and may be paid separately. DISMISSED.** Equation (3.12) is pointwise. The divergent debt in (3.14) is cancelled by the retained-square variance while the original packet has zero mean.

5. **Objection: R-087 already proves the sequential atom because the two atom sums have the same endpoint. VALID WITH CORRECTION.** The spatial conclusion does transfer, but not from endpoint equality: Theorem 4A.1 reruns the R-087 paraproduct proof on the identical three-channel sequential integrand. The totals agree by (6.4), while the termwise defect (6.7) has no sign. Neither argument proves the once-owned q_k^{seq} sum or the complete centred owner intertwiner.

6. **Objection: Sequential atomisation makes different physical shells orthogonal. DISMISSED.** The telescope occurs before the Hilbert–Schmidt square. Theorem 5.1 uses weighted Cauchy–Schwarz and assumes no cross-shell orthogonality.

7. **Objection: The rare-event fixture refutes the exact $\gamma = 7/12$ Cartan atom. DISMISSED.** It evaluates the extracted Z^6 majorant. R-091's saturated scalar current has the opposite, decaying scale on that fixture. The complete legally centred vector/multimode owner intertwiner and its q_k^{seq} sum remain open.

8. **Objection: Every fixed R-104 chart has a last block, so a refinement-uniform ellipticity constant may be chosen. DISMISSED.** Equation (9.4) sends the last-block covariance to zero as the final mesh interval shrinks. A fixed last interval is not cofinal for progressive controls, and adding or reordering roots changes the chart.

9. **Objection: Pointwise six-real ellipticity supplies all spatial data. DISMISSED.** The field (8.7) has covariance I_6 at every point and fractional norm of order N^{σ}. Conditioning can also make its pointwise lower eigenvalue zero as in (8.9).

10. **Objection: Theorem 5.1 closes controlled-shell one-use. UPHELD as an overclaim.** The uncentred production spatial atom is now Theorem 4A.1, but the legal centred conditioning step, once-owned q_k^{seq} sum, owner conversion, and exact R-125/R-088 intertwiner are not conclusions of Theorem 5.1.

11. **Objection: A sufficiently large collar always closes the final matrix. DISMISSED.** Equation (11.7) is a legal shrinking-headroom profile which defeats every collar. A positive h_* or one verified finite collar is separate data.

14. Remaining theorem, open gates, and no-overclaim statement

The Nelson-level target remains the complete-low R-123 signed trace-excess bound (2.9), equivalently (2.10). The smallest honest analytic successor toward that target is one of the following two production subtheorems.

1. Upgrade Theorem 4A.1 to the legally centred expectation-inside bound (5.3), with all output clusters, same-root visits, trace covariance, future variance, R-063 forest, rational and Cartan rows, low block, and owner conversion retained, and prove the once-owned sum (5.4).

2. Bypass per-atom bounds and estimate the complete variance-retained output ledger (5.5) directly, again without extracting a translated-model Z^6 factor or separating the forest-current cancellation.

Either route must use a fixed-chart aggregate or a shell-weighted joint value–gradient/increment/regression theorem; Theorem 9.1 rules out a common refinement-uniform last innovation. Afterward one still must certify the complete R-123 trace-excess inequality, near/balanced headroom (11.5), the low matrix, matching energy, strict gap, and absolute anchor before applying the R-129 collar mechanism.

The following remain open without qualification:

- `A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE` (T-050);
- `A13-CLASSII-FULL-PROGRESSIVE-REVISIT-EXTENSION`;
- the complete signed forest-current lower bound;
- the complete owner-to-projected-sequential-atom intertwiner;
- the legal centred-conditioning step and one-use q_k^{seq} ledger;
- $\mathrm{OVERLAP}_{\mathrm{src}}$ and the $q = 10/9$ Nelson bound;
- the interacting measure, cutoff/floor/volume removals, and Sector-A closure.

No claim is made that this note proves a production covariance lower bound, a cutoff/chart/control/refinement-uniform spatial estimate, a complete owner-intertwining theorem, a strict augmented gap, or either A13 gate. The directed-refinement theorem is a no-go for one proposed proof architecture; it is not a counterexample to the complete signed action or to Nelson.

15. Result footer

```
Result ID:              A13-CLASSII-VARIANCE-RETAINED-SEQUENTIAL-ATOM-
                        REFINEMENT-BOUNDARY (R-135).
Claim:                  A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-
                        OBSTRUCTION.
Precise statement:      The complete recombined owner is exactly one half
                        the adapted R-063 forest minus one half the
                        conditional future variance; the complete-low R-123
                        trace excess is the live aggregate target. The R-087 spatial proof
                        applies directly to the uncentred R-088 sequential
                        atom; at alpha=2/5 and s=2/3 its three margins are
                        1/15, 4/15, and 16/15. A centred sequential atom
                        hypothesis sums with square constant
                        0.658881625872614... at s=2/3 and gamma=7/12.
                        R-087 self atoms and R-088 sequential atoms have an
                        exact mixed-Hessian homotopy defect, so equality of
                        endpoint sums gives no termwise domination. The
                        extracted-Z6 route fails on the predictable rare
                        branch. No nonzero terminal source block can remain
                        independent and uniformly six-real elliptic over
                        the full directed temporal-refinement union because
                        its tail covariance tends to zero.
Scope:                 Fixed finite cutoff, positive floor, common legal
                        heat, one declared temporally faithful chart,
                        recombined same-root visits, and an exhaustive
                        contraction-closed orthogonal output family. The
                        refinement no-go quantifies over the R-093 directed
                        temporal union.
Dependencies:          R-087, R-088, R-091, R-092, R-104, R-116, R-119,
                        R-123, R-125, R-129, R-133, and R-134.
Evidence grade:        ANALYTIC/EXACT/EXECUTED with primary, non-importing
```

	independent, and integrated executable checks, a rendered PDF, visual QA, manifest, and release gate.
Reproduction command:	E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_variance_retained_sequential_atom_refinement_boundary_verify.py
Expected output:	R-135 integrated PASS: 142/142 integrated; aggregate 284/284, with primary 69/69 and non-importing independent 73/73 children PASS; PDF and generated-surface gates PASS.
Negative boundary:	The R-134 root-owner scalar covariance envelope is unbounded below on the scaled R-125 fixture, and refinement-uniform independent last-block ellipticity is impossible in the full directed temporal union. Both are method no-gos, not counterexamples to the complete production action. Authorities are NG-2026-07-31-A13-COVARIANCE-ENVELOPE-REBATE-ERASURE and NG-2026-07-31-A13-REFINEMENT-UNIFORM-LAST-BLOCK-ELLIPTICITY; the extracted-Z6 route reuses its 2026-07-25 no-go.
Falsification gate:	Any failed owner factor, variance identity, sequential telescope, homotopy formula, geometric constant, covariance-tail limit, or scope firewall.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No centred owner intertwiner, q-ledger, signed forest-current bound, headroom, A13 gate, Nelson theorem, measure theorem, removal limit, or Sector-A closure is proved.
Proof complete:	false. Sector A closed: false.
Next required action:	Prove the legal centred-conditioning/owner intertwiner and once-owned shell-weighted q-ledger, or a direct integrated substitute; prove the full R-123 trace-excess bound, then certify one finite collar's near/balanced headroom.